Modeling Human Responses to Multimodal AI Content

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We introduce the MhAIM Dataset, enabling large-scale analysis and model training of human responses. Our study shows that people are better at identifying artificial intelligence content when posts include both text and visuals, and propose two composite metrics for downstream risk assessment and decision support.

As artificial intelligence (AI) techniques rapidly advance, AI-generated content (AIGC) is becoming increasingly common in journalism, the arts, and social media.[1] While the benefits of AIGC are manifold, the potential harms cannot be overlooked. For instance, AI-crafted content, especially when multimodal in nature, can often be indistinguishable from genuine human-crafted content, leading to potential misinformation, deception, and erosion of trust in digital platforms.

While a plethora of studies have addressed misinformation and AI-generated content,[1] the current research primarily focuses on generating or detecting such content in isolation, or remains confined to a single modality. There is still a big gap in understanding how people react to and are influenced by multimodal AI-generated content. Addressing this gap is essential for fostering trust in online platforms, advancing digital forensics, and promoting the ethical use of AI.

To investigate the misinformation potential of AIGC, we present *T-Lens* (*Trust Lens*), an agent system that predicts human responses to online content and provides accessible, interpretable explanations (Figure 1). At its core is the human response–model context protocol (HR-MCP), a specialized module estimating human-centered attributes, such as trustworthiness and impact. To train and evaluate T-Lens, we constructed the Multimodal Human and AI Generated Media Dataset

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(MhAIM), comprising 154,552 posts produced by either humans or AI (see Figure 2 for examples). A human study on MhAIM reveals diverse reactions: AI-generated text is often seen as more trustworthy than AI visuals, and users are generally less inclined to believe or share content they suspect to be AI-generated. Yet, well-crafted AIGC can still be persuasive, even when its origin is recognized. These findings motivate the design of HR-MCP, which captures how people interpret and respond to multimodal content. Integrated into T-Lens, HR-MCP enhances transparency and helps mitigate the continued influence effect of misinformation.[2,3] Experiments on real-world data underscore the effectiveness of our approach.

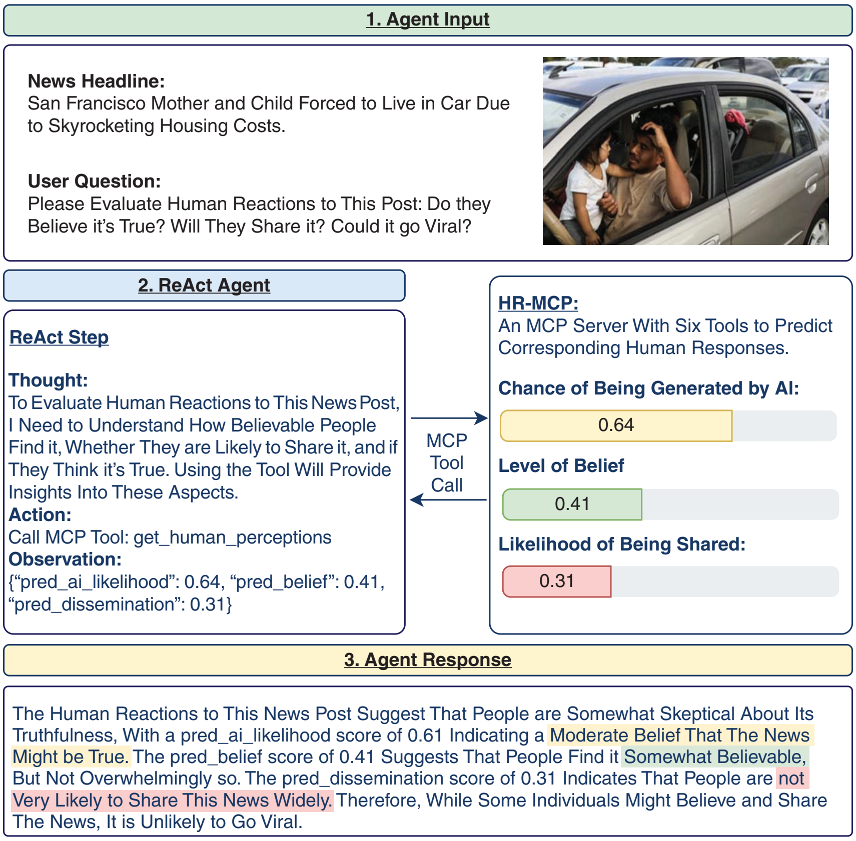

FIGURE 1. We study how users perceive AI-generated content and propose T-Lens, a large language model (LLM)-based agent that predicts human responses with interpretable explanations. At its core is human response model context protocol (HR-MCP), a module that reasons over both content and how humans are likely to interpret and respond to it.

We adopt a human-centric approach to address the challenges posed by AIGC and misinformation. Rather than focusing solely on detection, we prioritize predicting how people respond to content. This perspective is particularly important in domains, such as trading and the stock market, where anticipating user reactions (for example, whether a news post will go viral or influence investment behavior), can be more consequential than verifying its factual accuracy. Our human studies show that well-crafted AIGC can still appear credible and persuasive, highlighting the limitations of relying solely on detection. In contrast to methods that depend on external fact-checking, which are increasingly challenged by sophisticated generative tools,[3] T-Lens aligns with human perception

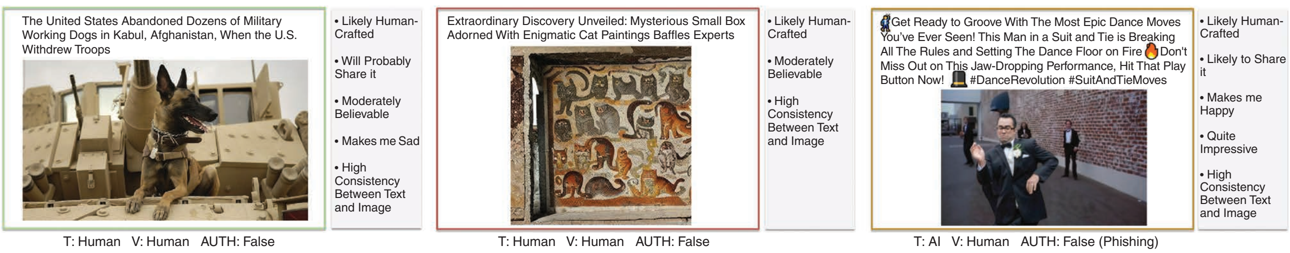

FIGURE 2. Examples from the MhAIM Dataset with origins of text (T) and visuals (V), authenticity (AUTH), and common human response shown. Unless noted, AI texts and visuals are generated by ChatGPT and Stable Diffusion, respectively.

and offers a more robust alternative. Our findings offer valuable insights for trading platforms, news aggregators, and social media systems aiming to strengthen automated review and reduce the impact of AI-driven misinformation. Our contributions are as follows:

- We present *T-Lens*, an agent system designed to generate human-aligned responses to multimodal content. The key module of T-Lens is HR-MCP, a specialized MCP server developed based on insights from our human study. HR-MCP enables the prediction of human responses to misinformation, capturing aspects, such as trustworthiness, impact, receptivity, user beliefs, and sharing tendencies. By integrating HR-MCP, T-Lens delivers accessible explanations that help mitigate the harmful effects of misinformation and address challenges posed by AIGC. Experimental results demonstrate that T-Lens significantly enhances the alignment of human responses produced by large language models (LLMs). Furthermore, HR-MCP follows a modular MCP standard, enabling seamless integration with various LLMs for human-centric reasoning.
- We carefully designed the human study and found that user sensitivity to AIGC varies, peaking for multimodal social media posts and decreasing for text-only content. Despite general distrust, well-crafted AIGC can still be influential. These findings provide the community with insights to develop targeted strategies against AI-driven misinformation.
- We construct MhAIM, a multimodal dataset of AI- and human-generated content. A subset of MhAIM has been annotated with human responses, including elicited emotions, perceived credibility, and the likelihood of sharing. MhAIM supports understanding human-AIGC interaction and aids in both countermeasures against malicious use of AIGC and strategies for its positive application.

RELATED WORK

Multimodal analysis on misinformation

Multimodal analysis on misinformation leverages text, image, and video interactions, but this approach has not yet been commonly applied to AIGC. There is growing interest in detecting AI-generated content,[1,4] using methods like deep learning for text and neural networks for AI-synthesized imagery, including deepfake videos and real-time deepfakes (see review[4]); however, they are mainly in single modality. We found that well-crafted (that is, highly realistic within a single modality) AIGC can still gain trust, so solely detecting it may not capture human responses adequately. Hence, we prioritize multimodal consistency, aligning with human perception and receptivity.

Model context protocol

The integration of external tools into LLMs has advanced rapidly. OpenAI's 2023 function calling mechanism allowed models to interact with external functions,[5] but required manual interface definitions and authentication, limiting scalability. To address this, Anthropic introduced the MCP in 2024, providing a unified framework for tool interoperability. MCP uses a standardized JSON-RPC 2.0 protocol, enabling seamless communication between hosts, clients, and servers via standard input/output and server sent event channels. By formalizing roles for hosts (AI applications), clients (connectors), and servers (capability providers), MCP reduces engineering overhead and supports modular AI agent deployment.[6] Building on MCP, we developed T-Lens, a modular agent system that combines human-centered reasoning with dynamic tool integration to predict human responses to multimodal content.

DATASET AND HUMAN STUDY

We constructed the MhAIM Dataset, which consists of 111,153 pieces of AI-generated content and 43,309 pieces of human-crafted materials (see examples in Figure 2).

Stimuli collection and AIGC generation

To build MhAIM, we integrated data from multimodal misinformation and authentic sources, including Fakeddit,[7] N24News,[8] and FaceForensics++.[9] This diverse integration enables MhAIM to capture the complexity of both real and misleading content. See Supplementary Section 1.1 for details (see the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors).

We generated AIGC based on aforementioned real-world data. Text prompts from our collection were used with Stable Diffusion XL 1.0 to produce images, and an AI Headline Generator, built using LangChain and ChatGPT with tools like VQA, produced diverse AI texts across styles and media types. We also used the Gemini application programming interface to create texts from original and AI-generated visuals. In total, we constructed 111,153 AI-generated entries. Details and examples are in Supplemental Sections 1.1–1.2 (see the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors).

Human psychophysics study

We conducted an extensive human study using the MhAIM Dataset. We recruited 765 participants (ages between 19 and 65 years, 390 females) from the crowdsourcing platform Toluna, spanning three countries (participant numbers in parentheses): the United States (382), India (193), and Singapore (190). We further recruited 20 students (ages between 19 and 34 years, 12 females) from a local university. All participants received compensation for their involvement in the study. Participants viewed 30 posts (single or multimodal) and completed a questionnaire after each, assessing emotional response, behavioral tendencies (for example, belief, sharing), perceived AI origin, and modality consistency (see Supplementary Table S4 in the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors). Question phrasing varied by content type to capture how users perceive and engage with AIGC versus human content.

Due to budget constraints, our human study concentrated on a subset of MhAIM, encompassing 3,074 multimodal posts from different origin (AI-generated or human-crafted) and type (news, social media posts, and phishing messages). Each post received annotations from three to five participants, yielding 8,952 total human annotations. Detailed study design, distribution, questionnaire, participants' demographics, and an overview of human-annotated data can be found in Supplemental Sections 1.3–1.4 (see the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors).

Key findings from human study

Definition and method. We assessed participants' ability to distinguish AI-generated from human-crafted content by asking them to rate the "AIGC likelihood" of each post. Following signal detection theory, we computed sensitivity d' and bias c' as in prior misinformation work.[10] Sensitivity (d') is on an unbounded scale (higher values indicate better discrimination); it measures how well participants distinguish AI ("signal") from human ("noise") content. Bias (c') reflects response tendency: $c' = 0$ indicates no bias, $c' > 0$ favors human content, and $c' < 0$ favors AI content.

We employed inferential statistics to examine differences in the aforementioned metrics, using standard tests in behavioral sciences. We evaluated whether each metric varied by multiple factors, initially using ANOVA to check for general effects, followed by post hoc Tukey tests for significant ANOVAs to pinpoint manipulation effects, with a Bonferroni correction applied for multiple comparisons, setting a reduced alpha of 0.01. Below we report five major observations. Additional findings and detailed statistics with visualizations are reported in Supplemental Section 2 (see the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors).

Human sensitivity toward AIGC. *Observation 1:* Participants' sensitivity toward AIGC varies based on modality and information type. Our participants exhibited the highest sensitivity for multimodal social media posts and the lowest for social media texts without accompanying visuals.

On average, participants did slightly better than random guessing ($d' = 1.64 \pm 0.38$) in distinguishing AIGC from human-crafted content. The type of information they were judging made a big difference to their sensitivity, $F(4,\ 5{,}287) < 61.68$, $p \pm 0.00001$.[a] As shown in Figure 3(a), our participants were best at spotting AI content when it included both text and visuals on social media ($d' = 2.83$), but had a harder time with text-only posts ($d' = 0.87$). Participants tended to be cautious during judgments, leaning toward classifying content as human-crafted [suggested by positive bias in Figure 3(b)]. The large gap between multimodal social media ($d' = 2.83$) and text-only posts ($d' = 0.87$) may come from multiple factors: Current generative systems are often stronger in text fluency than in image realism, so visual artifacts or text–image inconsistencies can provide additional cues in multimodal posts. It may also reflect human reliance

[a]ANOVA results are reported as "$F(\mathrm{df}_{\mathrm{condition}}, \mathrm{df}_{\mathrm{error}}) = F$ value, $P = P$ value," with $P < 0.05$ rejecting the null hypothesis.

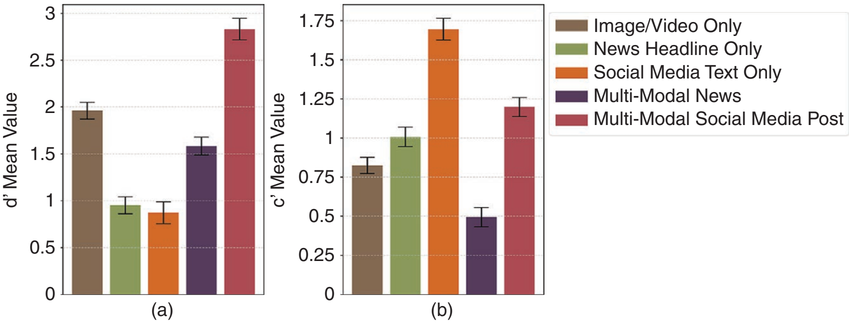

FIGURE 3. Users are most accurate at recognizing AI-generated multimodal social media posts and least accurate with AI-generated text (a). They tend to classify content as human-crafted, especially for social media posts, as indicated by the positive bias (c') (b). Error bars represent standard errors, showing variations among individuals.

on visual evidence during credibility judgments; both mechanisms are likely to contribute.

Observation 2: Mismatches between text and the corresponding visuals generally lead to the identification of posts as AI-generated. Specifically for news posts, the perceived veracity of the content plays a pivotal role.

As shown in Table 1, participants frequently identified multimodal content as AIGC due to mismatches between text and visuals. Specifically, for news content, perceived information credibility was a crucial factor.

Receptivity and trust in AIGC. *Observation 3:* AI text is more well-received than AI visuals, while human-crafted content, regardless of whether it's real or fake, consistently gets higher receptivity than AI-generated content.

Besides having participants assess whether a post was AI-generated ("AIGC likelihood"), we also gauged their level of belief in the content's veracity ("belief") and their inclination to share it ("dissemination propensity"). Based on these ratings, we develop two novel metrics to evaluate human receptivity to AIGC[b]: *trustworthiness*, indicating user trust, calculated as the difference between belief and AIGC likelihood (trustworthiness = belief – AIGC likelihood); and *impact*, measuring effect on individuals and the community, computed as the sum of belief and dissemination propensity (impact = belief + dissemination propensity). Both metrics are normalized to the range [0, 1] using min-max normalization.

Our human-crafted posts, whether real or fake, consistently exceeded AI-generated content in both composite metrics. AI text significantly outperformed AI visuals in impact ($F \geq 28$, $p \leq 0.0001$), a disparity likely due to the more developed capabilities of AI in text generation compared to visual content creation. The impact of AIGC increased alongside stronger perceived positive sentiments ($\rho \geq 0.6$, $p \leq 0.0001$) and greater multimodal consistency ($\rho \geq 0.54$, $p \leq 0.0001$), underscoring the importance of these elements in influencing

[b]In our study, these metrics also apply to human-crafted content, as we present information to users without explicitly labeling it as AI or non-AI.

TABLE 1. Participants' reasons for classifying multimodal information as AIGC or human-crafted.

	Cues for AI origin	Cues for human origin
News	The news seems not true.	The news seems true to me.
	Texts and visuals mismatch in contents or sentiments.	Texts and visuals align in content and sentiment.
	The visual may be AI-made, but the text isn't.	The style is typical of news outlets
Social media post	Texts and visuals mismatch in contents or sentiments.	Texts and visuals align in content and sentiment.
	The content doesn't make sense.	The style akin to common social media platforms.
	The visual may be AI-made, but the text isn't.	I have seen a similar post before.

Content veracity (brown text) and text-visual mismatch (blue text) are top reasons for news and social media posts, respectively.

user acceptance. Visualizations are in supplementary material Figures S11 and S12 (see the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors).

Observation 4: As people believe information is more likely to be created by AI, their receptivity toward it decreases. However, well-crafted AIGC can still have a significant impact even when people perceive it to be AI-generated.

People were generally less likely to believe or share content perceived as AI-generated ($Fs \geq 13.98$, $ps \leq 0.0001$;) (Figure 4). However, exceptions existed. We compared 50 highly likely AI-generated posts with high sharing intent (average share = 0.65) to 50 similarly perceived posts with low sharing (average = 0.05). Despite similar AI likelihood (~0.90), the more shareable group had stronger text–visual consistency, emotional impact, and perceived trustworthiness (see Supplementary Figure S14 in the downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors). This suggests that well-crafted AIGC can still gain traction, even when recognized as AI-generated.

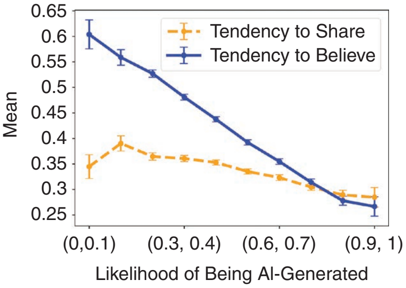

FIGURE 4. As participants grew more convinced a post was AI-generated, they were less willing to trust and share it.

Discussion. Our human study revealed relatively low human sensitivity to AIGC. This aligns with challenges identified in prior work on foundation models[4] and highlights the importance of multimodal consistency. Mismatched text and visuals often led to perceptions of AIGC, reducing belief and sharing intent. However, well-crafted AIGC showed high receptivity, even when users suspect its AI origin. These findings are based on current state-of-the-art models and may evolve with future AIGC quality. Our results are influenced by user backgrounds, with error bars representing human variance (see Supplementary Section 2 for analysis of the individual differences, in the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors). This human study contributes to discussions on AIGC's misinformation potential and inspired the development of T-Lens, our model that predicts human reactions to digital information.

FRAMEWORK DESIGN

We introduce T-Lens, a ReAct-style agent system designed to reason not only over content but also over how humans are likely to perceive and respond to multimodal information (see Figure 5). At the core of T-Lens is the HR-MCP, a dedicated MCP server trained to predict five key types of human responses to misinformation. Guided by insights from our human study, HR-MCP enables the agent to align its reasoning with how users interpret and emotionally react to multimodal content.

Overview of T-Lens framework

T-Lens is built on the ReAct agent[18] and augmented with HR-MCP tools to support nuanced reasoning over user prompts and multimodal inputs (text and images). For each task, T-Lens performs up to 15 reasoning steps (configurable based on complexity), during which the LLM determines whether a human-like interpretation is needed. If so, it invokes one of the HR-MCP tools. Each step in T-Lens includes a *thought* (rationale for calling an mcp tool), an *action* (mcp tool name and parameters), and an *observation* (mcp tool output). This design allows the agent to iteratively gather relevant semantic, emotional, and contextual cues before producing a final user-facing response. T-Lens supports interfaces with GPT-4o and Gemini, and is also compatible with open-source models like LLaMA, Qwen, and DeepSeek. In this article, we used GPT-4o, GPT-4o, one of the leading and widely-used LLMs, as the agent's backbone. A detailed prompt example and a screenshot are shown in Supplementary Figures S15 and S16 (see the supplementary downloadable material available at https://doi.org/10.1109/MC.2026.3683417, provided by the authors).

Design of HR-MCP

HR-MCP is an MCP server integrated within T-Lens to model human responses to multimodal content. Since AIGC often combines text and images, we adopted a CLIP-style architecture with a vision transformer (ViT)-based image encoder and a BERT-style text encoder. The model was trained on human-labeled data, making it well-aligned with identified human responses to multimodal information.

Multimodal semantics consistency. Through our human study, we found that discrepancies between text and visual significantly influenced the classification of content as AI-generated. Perceived content veracity emerged as the primary factor for AI news detection. Furthermore, the likelihood of content being AI-generated significantly impacted human receptivity. Drawing from these insights, the initial step of HR-MCP is to compute information semantics and model multimodal semantic consistency. To achieve this, we used a ViT[19] to obtain image embeddings and a BERT-like architecture with 12 layers of transformer blocks, each with a hidden size of 768 and 12 self-attention heads, to compute text embeddings. We use 12 layers following the default CLIP text-encoder configuration used for initialization, so the architecture remains consistent with the pretrained backbone. Specifically, we utilized the base version of ViT with a 32×32 patch size, taking a $224 \times 224 \times 3$ image as input and outputting a 512-dimensional image embedding. For text encoding, the encoder handles tokenized sentences, resulting in a 512-dimensional text embedding. Concatenating these produces a 1,024-dimensional multimodality embedding, encapsulating the semantics features from each modality as well as text-visual consistency. By this, HR-MCP learns key factors impacting AIGC perception and human receptivity, as identified in our human study.

Multimodal sentiment consistency. Our human study found that emotional mismatches between text and visuals indicate AIGC likelihood. Hence, after extracting multimodal embeddings, we added a sentiment module to compute emotion consistency between text and corresponding visuals. The module, an MLP-style subnetwork, takes a 1,024-dimensional raw multimodality embedding as input. The subnetwork consists of three linear layers (1,024, 2,048, and 1,024 nodes), uses ReLU activation, and includes a dropout layer (rate = 0.5, PyTorch default rate) to encourage convergence and prevent overfitting. The sentiment module outputs a 1D feature representing sentiment consistency between text and visuals.

Embedding fusion. Obtaining two embeddings (semantics and sentiment) from previous steps, we summed them and applied Layer Norm for smoother gradients and convergence, yielding a 1,024-dimensional

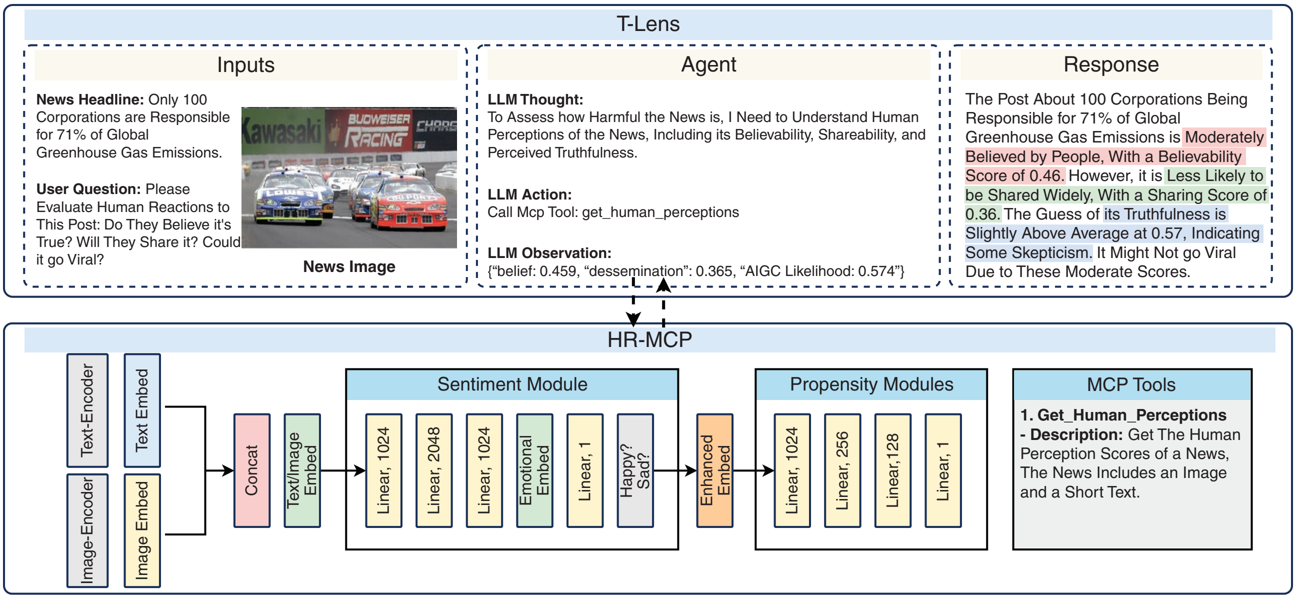

FIGURE 5. T-Lens framework: Predicting human responses through ReAct-style reasoning with integrated HR-MCP. Given multimodal inputs (text and image), the agent follows ReAct steps: *Thought*, *Action*, and *Observation*. At each step, it determines whether human response signals (for example, belief, dissemination tendency) are needed. If so, it queries HR-MCP, which returns predicted scores to guide the agent's reasoning and generate explanations grounded in human reactions. HR-MCP is designed as a plug-and-play module, compatible with any LLM, making it easy to integrate into a wide range of agent systems.

enhanced embedding. This ultimate embedding encompasses image and text content, along with potential consistency between text and visuals in terms of both content and emotions.

The final submodule contains three identical propensity modules to predict three human responses (AIGC likelihood, belief, dissemination propensity) and computes two composite metrics (trustworthiness and impact). Each employs an MLP with four linear layers (1,024, 256, 128, and 1 nodes), using ReLU activation and dropout (rate = 0.5) after each layer.

EXPERIMENTS

Experimental settings

The HR-MCP model parameters for both the image encoder and text encoder were initialized using the pretrained CLIP model.[20] Throughout the training process, we used a joint mean squared error loss of the predicted information attributes (AI likelihood, belief, dissemination). The parameters of the deep neural network model are learned in an end-to-end manner, leveraging the Adam optimizer with a learning rate of 0.0005. We used this fixed learning rate as the default training setting in our implementation, and no learning-rate scheduler was applied. We trained the model for 50 epochs with a batch size of 64. The complete training procedure takes approximately 0.5 h on a single NVIDIA 3090Ti GPU and the PyTorch framework. We randomly split 3,074 multimodal posts (with 8,952 total annotations) into 80% as a training set and 20% as a test set at the post level.

Evaluation metrics. We used Spearman's correlation (ρ), a standard metric for modeling human reactions, and area under the curve (AUC) to capture binary decision-making behavior (for example, whether to share a post). Since decisions are often binary, we binarized annotations at a 0.5 threshold; for example, we argue that a predicted 0.8 better matches a ground truth of 0.6 than 0.4.

As no existing models directly target human receptivity like T-Lens, we compared it with four multimodal misinformation detectors: FNR,[14] CAFE,[13] MCAN,[12] and the widely used EANN.[11] We excluded some recent models due to dataset or preprocessing limitations. All models were trained with consistent fivefold cross-validation, and results were averaged over five runs. Because these detectors are originally designed for classification (for example , real versus fake), we replaced their final classification layers with regression heads to predict the continuous human response attributes (belief, dissemination, etc.) and finetuned them under the same protocol as T-Lens for fair comparison. More importantly, we also compared T-Lens with recent powerful multimodal LLMs (MLLMs), including GPT-4o,[15] Gemini 2.5 Flash,[16] and Claude 3.7 Sonnet,[17] applied without additional tuning. This allowed us to assess T-Lens against top-tier and widely used MLLMs in modeling human receptivity.

Experimental results

Table 2 shows that T-Lens consistently outperforms all compared methods and MLLMs across the five metrics. T-Lens excels in predicting participants' AIGC likelihood and belief (ρs $\geq$ 0.400, AUCs $\geq$ 0.701) and effectively assesses information trustworthiness and impact. T-Lens is less effective in predicting dissemination (ρ = 0.284, AUC = 0.661), likely due to the complex nature of sharing decisions, which are influenced by factors, such as a user's background (for example, political bias, cultural preferences, etc.) and online behavior. Overall, our model demonstrates the predictive power in both regression and binary classification tasks, showing the ability in predicting human responses to digital information. The modest performance of the comparison methods underscores their design for disparate purposes, such as multimodal misinformation detection, rather than their comprehensive efficacy. The superior performance over MLLMs highlights the contribution of HR-MCP, which is specifically designed to model human responses to online information.

Ablation study. We compared our method against three baselines to evaluate the impact of our proposed modules on simulating human responses. The baselines included a model with only text, a model with only visuals, and a base T-Lens model without the sentiment module (that is, we removed the MLP-style sentiment subnetwork; the CLIP-style encoders and their tokens were unchanged). As shown in Table 2, T-Lens outperforms all baselines in both regression and binary classification tasks, demonstrating the superiority of combining multimodality. Notably, the sentiment module significantly bolsters performance, mirroring our human study's finding on the role of sentiment consistency. This finding highlights the importance of incorporating affective dimensions into computational models for a more comprehensive understanding of human–AI interactions. Our experimental results also align with previous research, which emphasizes that affective dimensions play a crucial role in misinformation.[2]

In summary, our experiment suggests that, by incorporating both textual and visual, as well as the sentiment module, T-Lens effectively aligns with human cognitive processes, providing a more accurate simulation of how users interact with digital information. This comprehensive approach ensures that the model can better predict human responses, making it a valuable tool for combating misinformation in a multifaceted media landscape.

Generalizability and application. To evaluate T-Lens' generalizability, we directly tested it without fine-tuning on another dataset, Mediaeval2015, with 1,022 multimodal tweets and their corresponding retweet counts, akin to dissemination propensity in our case. Our model achieved a ρ of 0.270 in predicting retweet counts, close to the results on MhAIM, suggesting promising potential for T-Lens' applicability to real-world scenarios.

Discussion and future work. Experiments show T-Lens's potential, particularly its ability to infer trustworthiness by prioritizing multimodal consistency over AIGC detection, aligning with human perception. Unlike models relying on external verification, which can be unreliable with the rise of AI-generated misinformation, T-Lens leverages insights from our human study, mirroring real-world behavior where people often don't cross-verify information. While T-Lens demonstrates promising results, future research could enhance its capabilities by incorporating individual behavioral traits that influence receptivity to misinformation, thus providing a more comprehensive understanding of the human–AI–information interaction landscape.

In this work, we studied and model human responses to AI-generated content to mitigate misinformation in its increasing prevalence. We

TABLE 2. Merged results: Conventional models, LLMs, and ablation settings evaluated on five human response attributes with Spearman's ρ and AUC.

	AI likelihood		Belief		Dissemination		Trustworthiness		Impact	
Model	ρ	AUC	ρ	AUC	ρ	AUC	ρ	AUC	ρ	AUC
Comparison with conventional models and MLLMs										
EANN[11]	0.171	0.595	0.165	0.581	0.111	0.551	0.23	0.625	0.162	0.557
MCAN[12]	0.219	0.6	0.26	0.635	0.198	0.604	0.226	0.62	0.248	0.604
CAFE[13]	0.078	0.548	0.109	0.559	0.119	0.582	0.108	0.543	0.157	0.588
FNR[14]	0.179	0.613	0.195	0.614	0.14	0.587	0.244	0.654	0.191	0.599
GPT-4o[15]	0.314	0.679	0.333	0.667	0.132	0.622	0.353	0.685	0.251	0.595
Gemini-2.5-Flash[16]	0.337	0.689	0.34	0.667	0.098	0.562	0.371	0.749	0.252	0.636
Claude-3.7-Sonnet[17]	0.245	0.621	0.276	0.645	0.164	0.658	0.31	0.6	0.235	0.581
Ablation study										
Baseline: GPT-4o + HR-MCP (without visuals)	0.145	0.575	0.186	0.603	0.179	0.601	0.195	0.724	0.23	0.621
Baseline: GPT-4o + HR-MCP (without text)	0.278	0.634	0.233	0.62	0.185	0.609	0.317	0.682	0.205	0.594
Baseline: GPT-4o + HR-MCP (without sentiment)	0.307	0.645	0.294	0.663	0.203	0.628	0.343	0.698	0.27	0.641
T-Lens (ours, GPT-4o + HR-MCP)	0.414	0.714	0.400	0.701	0.284	0.661	0.375	0.751	0.293	0.717

introduce the MhAIM Dataset, comprising 111,153 pieces of AI-generated and 43,309 pieces of human-crafted multimodal information. We introduce T-Lens, an agent system that predicts how people will respond to multimodal information with reasoned explanations. The component of T-Lens is HR-MCP, which uses the standardized MCP, allowing it to seamlessly integrate into any LLM to enhance tasks that require modeling human responses. This research represents one of the first attempts to evaluate generative AI's misinformation potential from a human-centric, multimodal viewpoint.

Regarding limitations and future directions, our human study recruited participants from three countries (the United States, India, and Singapore). Although this covers diverse media environments, the findings may not generalize to all regions; therefore, claims should be interpreted with appropriate caution rather than as globally representative. In addition, future versions of HR-MCP can be improved by explicitly modeling individual behavioral traits that may shape misinformation receptivity, such as cognitive reflection, political leaning, media literacy, and prior exposure to AIGC tools.

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